Name: Kyle Salgado-Gouker

Date: February 29, 2024

Class: DSC 550 – Professor Werner

Project Milestone #4

Research Paper

Introduction

In the realm of political decision-making, understanding the factors that influence voting patterns is crucial for gauging the passage of bills. The purpose of this data science project revolves around predicting the voting tendencies of representatives and senators on legislative bills, specifically whether they will vote "Aye" or "Not Aye" (which does not just include "nay," but also "present" or "abstain" votes). By analyzing these voting trends, this project aim to forecast the outcome of bills and delve into the role of financial influence on legislative decisions.

The importance of this endeavor stems from its potential to enhance transparency and comprehension within the legislative process. Predicting voting outcomes can provide valuable insights into political dynamics, aiding both policymakers and the public in understanding the underlying factors driving legislative decisions. Moreover, by scrutinizing the correlation between financial contributions and voting patterns, we strive to shed light on the nuanced relationship between money and political influence.

To gain buy-in from stakeholders, it's pivotal to highlight the project's relevance and potential impact. By framing the problem as a means to enhance understanding of legislative dynamics and promote transparency within government processes, stakeholders can appreciate the project's significance in fostering informed decision-making and accountability. Emphasizing the predictive aspect underscores its utility in anticipating legislative outcomes, which can empower stakeholders to make more informed strategic decisions.

The project employs advanced data science techniques that analyze voting patterns as well as discern underlying trends. Machine learning algorithms are leveraged to predict the likelihood of representatives and senators voting in favor or against bills based on historical voting records and associated contextual factors. Additionally, statistical analysis is conducted to examine the impact of financial contributions on voting behavior, elucidating the intricate dynamics between money and legislative decision-making.

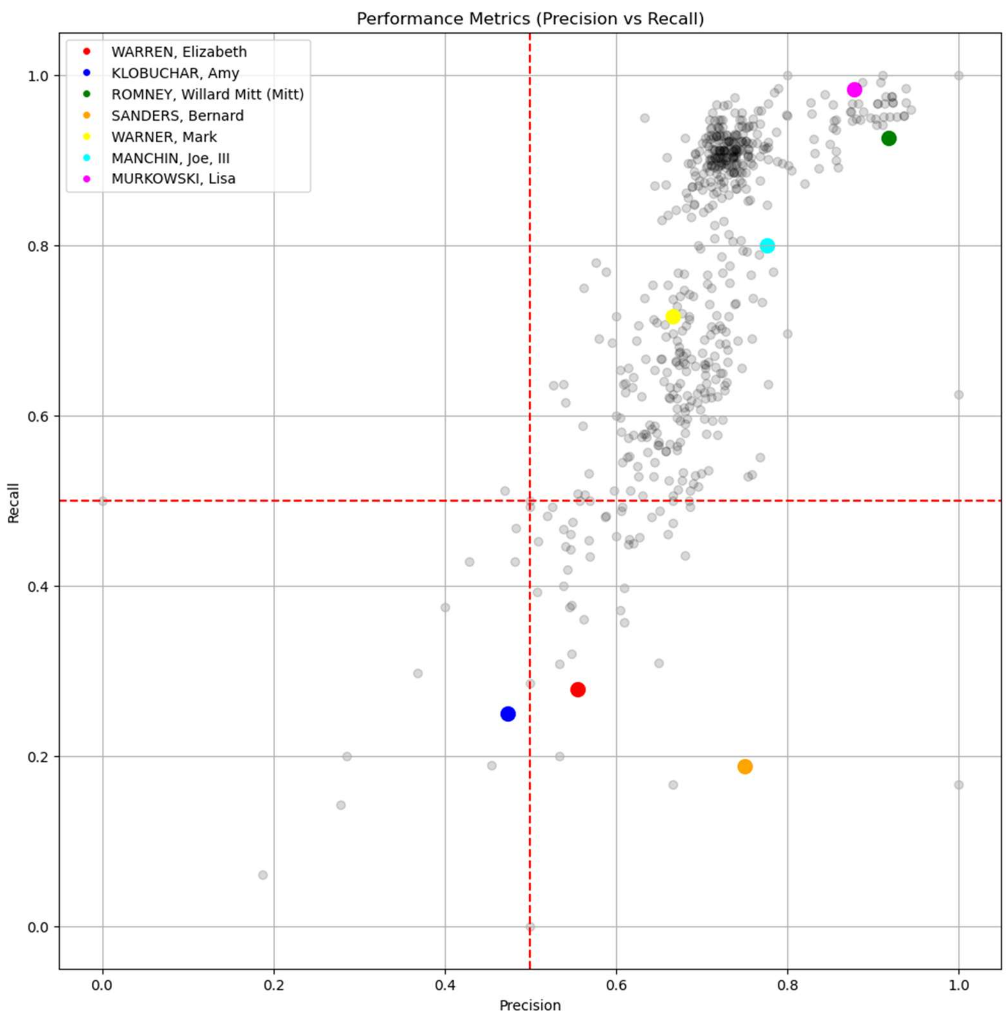
Graphical representations serve as powerful tools for visualizing insights gleaned from the data. Through interactive graphs and visualizations, key trends and patterns are elucidated, facilitating intuitive comprehension and interpretation of the findings. By presenting data-driven insights in a visually compelling manner, stakeholders are empowered to derive actionable insights and make informed decisions based on empirical evidence

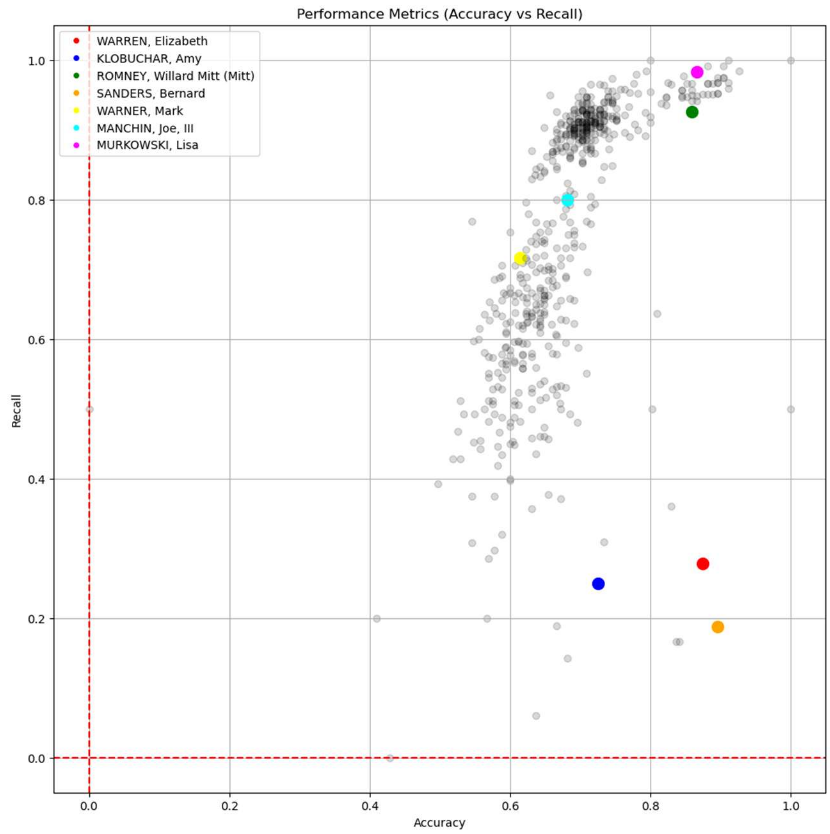
The data utilized for this project is sourced from comprehensive datasets encompassing legislative voting records and campaign finance disclosures. These datasets offer a rich repository of information spanning various legislative sessions and encompassing a diverse array of bills and political contributions. By leveraging this extensive dataset, we aim to capture the multifaceted interplay between legislative voting behavior and financial influence.

In summary, this data science project endeavors to unravel the complexities of legislative decision-making by predicting voting outcomes and exploring the role of financial influence therein. By harnessing the power of data analytics and machine learning, we aspire to enhance transparency, accountability, and understanding within the realm of political governance.

Through collaborative engagement with stakeholders, we endeavor to foster an environment conducive to informed decision-making and democratic governance.

Milestones Summary

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KLOBUCHAR, Amy
ROMNEY, Willard Mitt (Mitt)
SANDERS, Bernard
WARNER, Mark
MANCHIN, Joe, III
MURKOWSKI, Lisa
Recall
Accuracy
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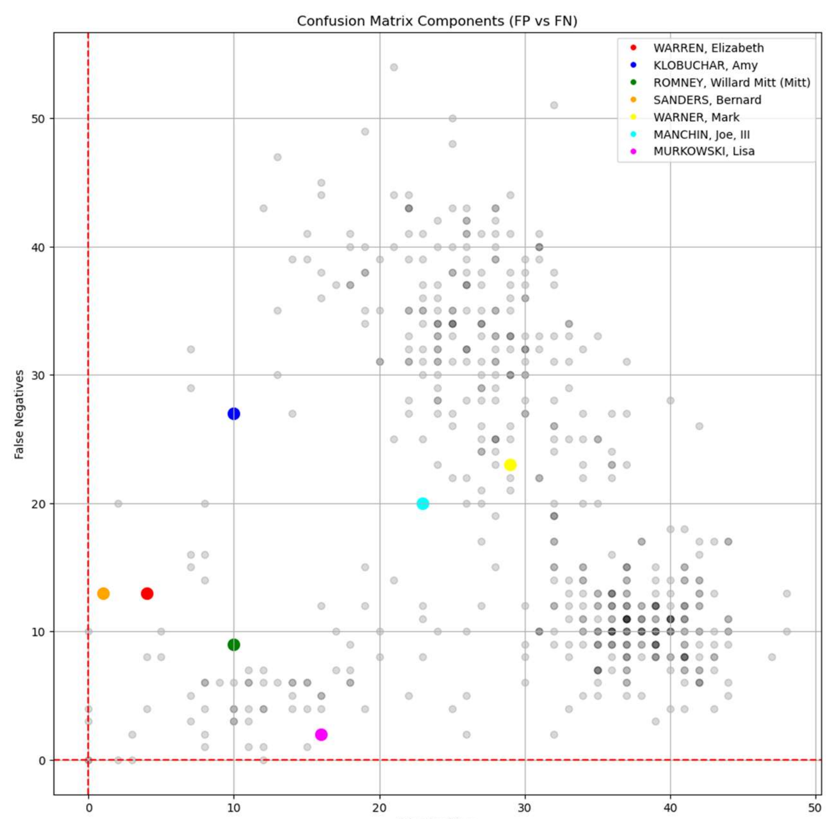
0.4
0.6
0.8
1.0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KLOBUCHAR, Amy
ROMNEY, Willard Mitt (Mitt)
SANDERS, Bernard
WARNER, Mark
MANCHIN, Joe, III
MURKOWSKI, Lisa
False Negatives
False Positives
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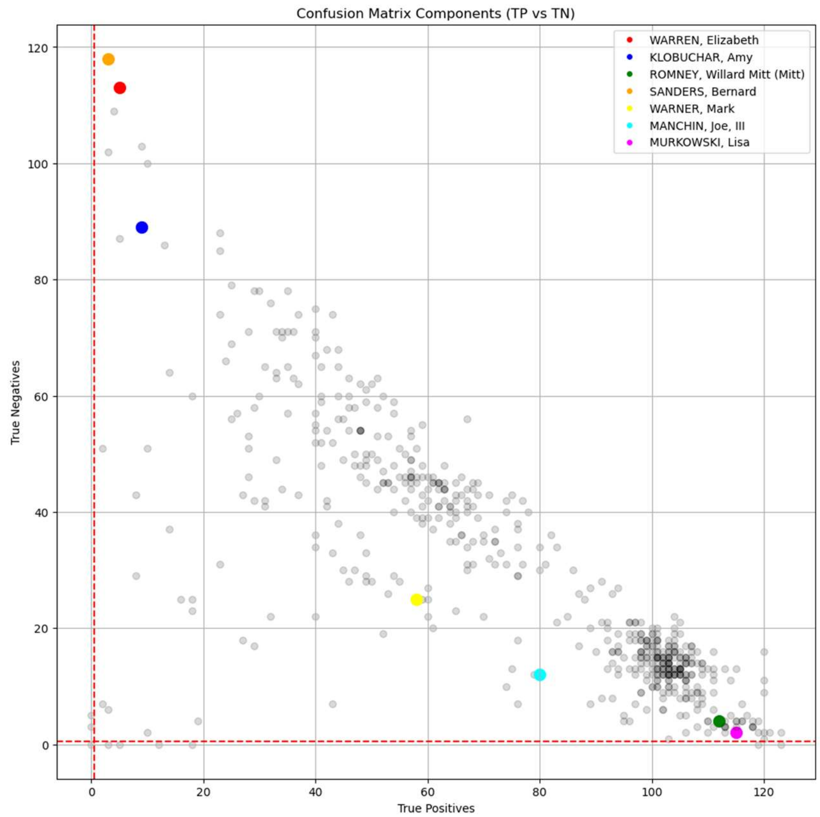
10
20
30
40
50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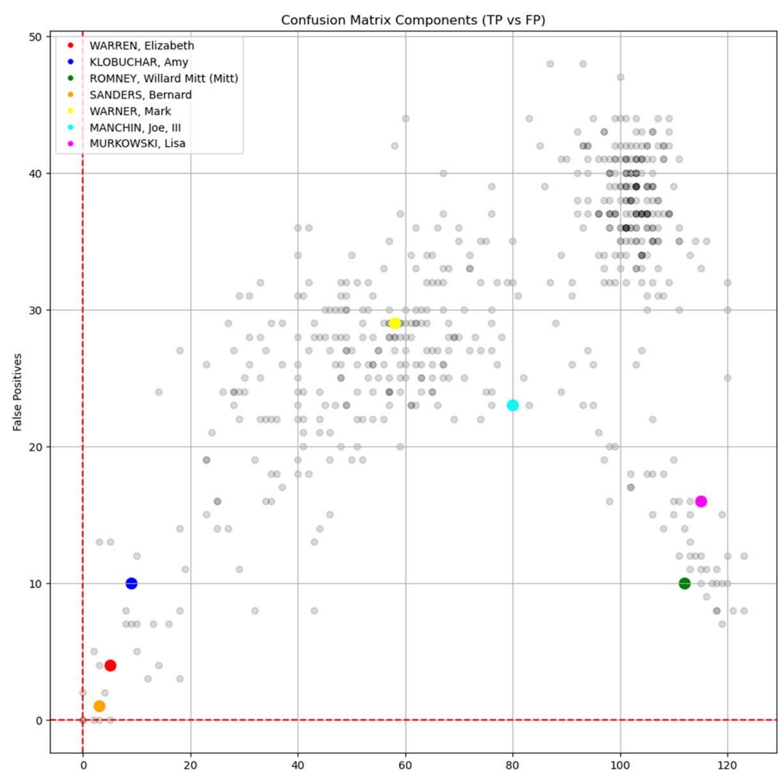
MURKOWSKI, Lisa
True Negatives
True Positives
0
20
40
60
80
100
120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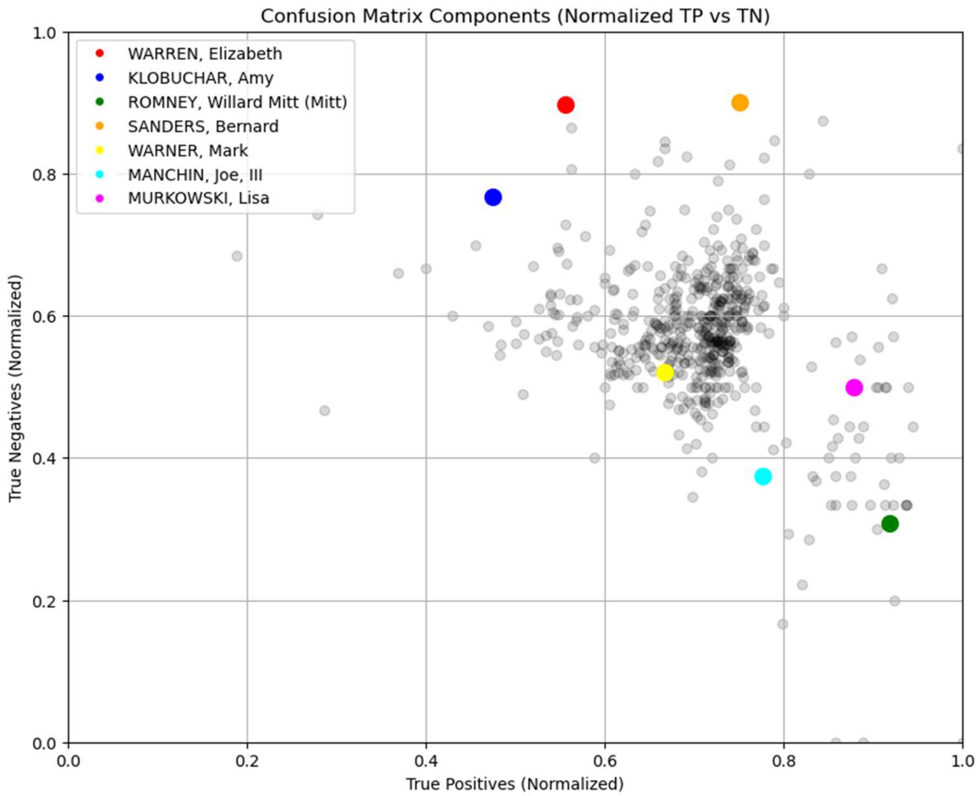
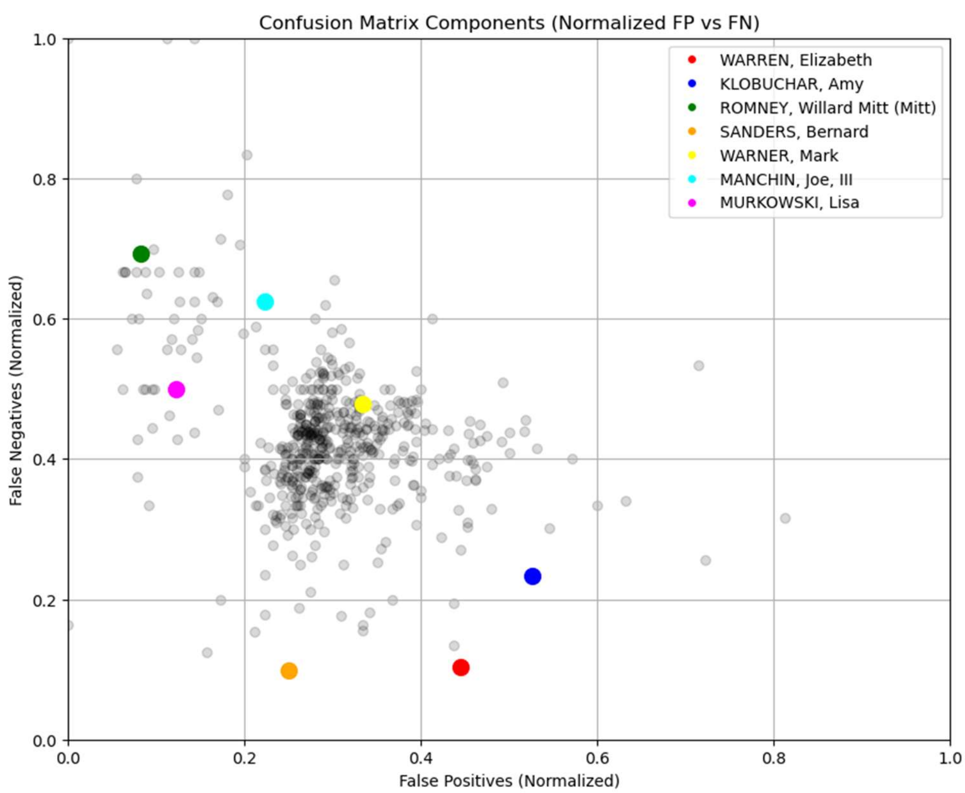
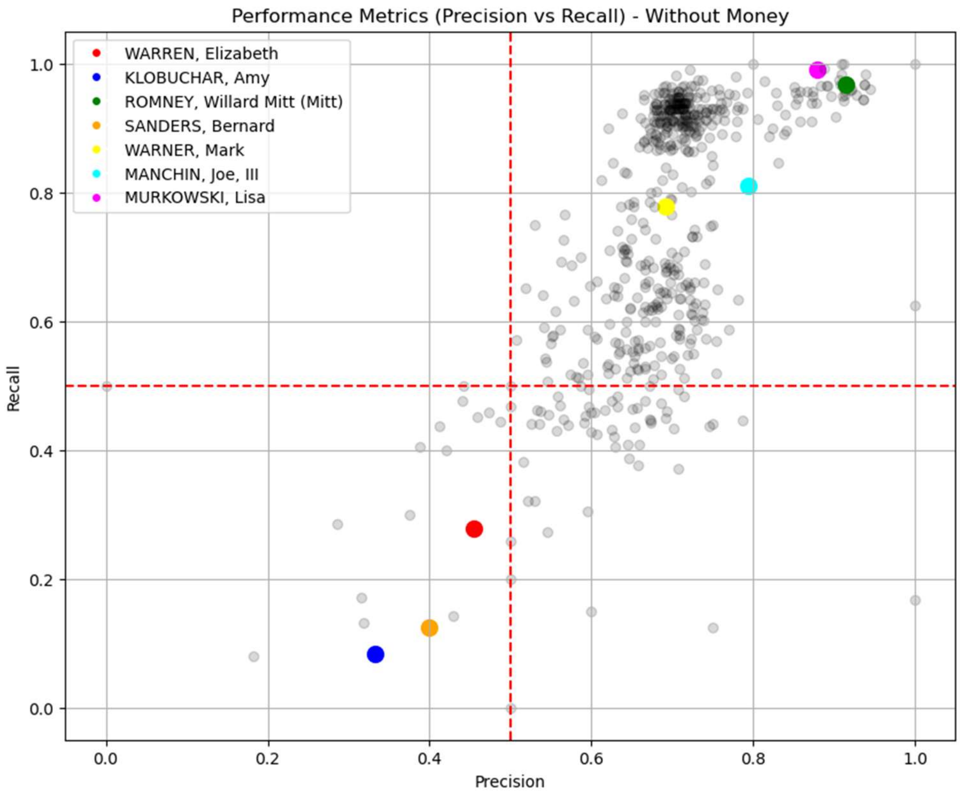
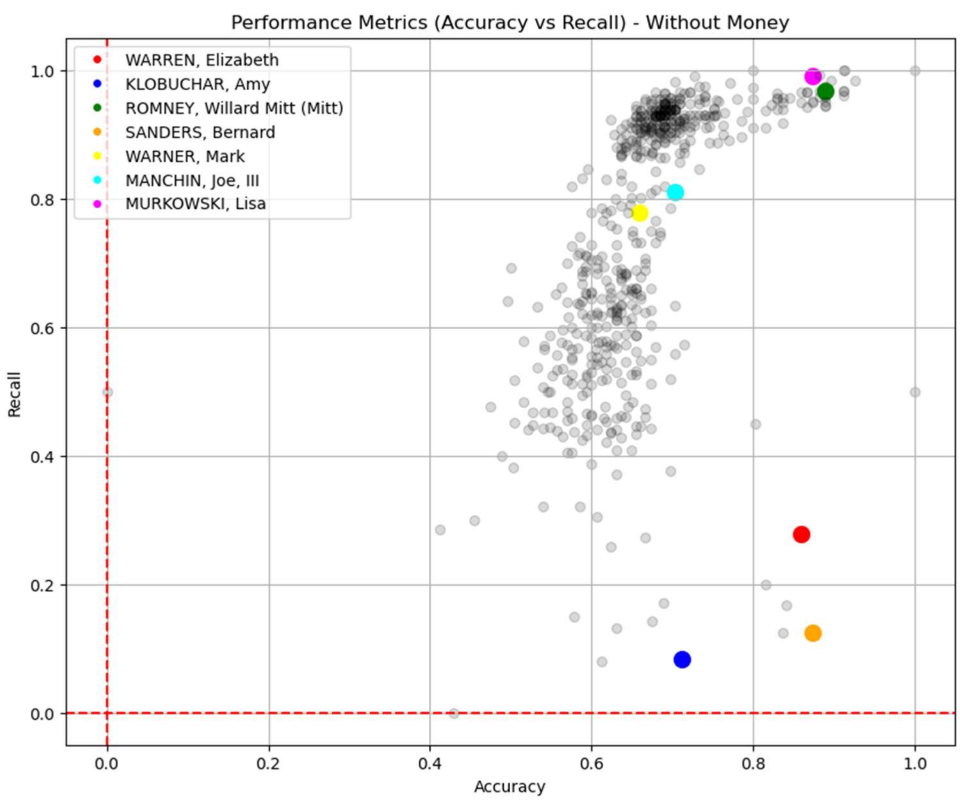
Normalized Graphs

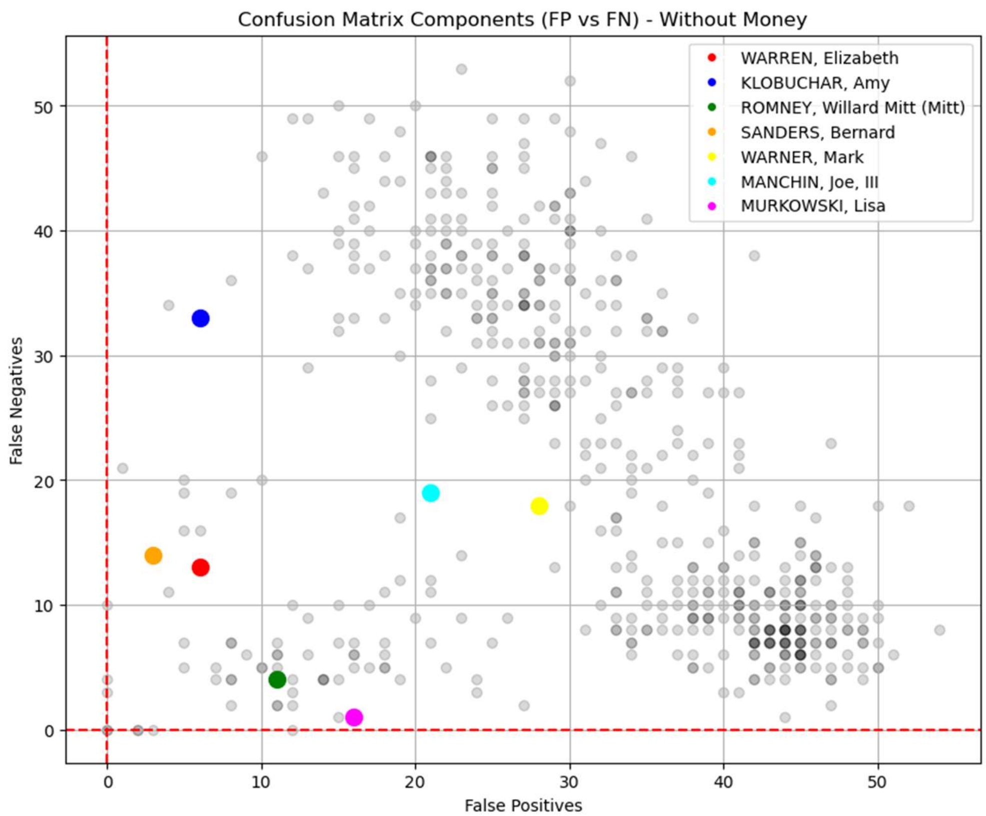




Without Money?






Confusion Matrix Components (FP vs FN) - Without Money
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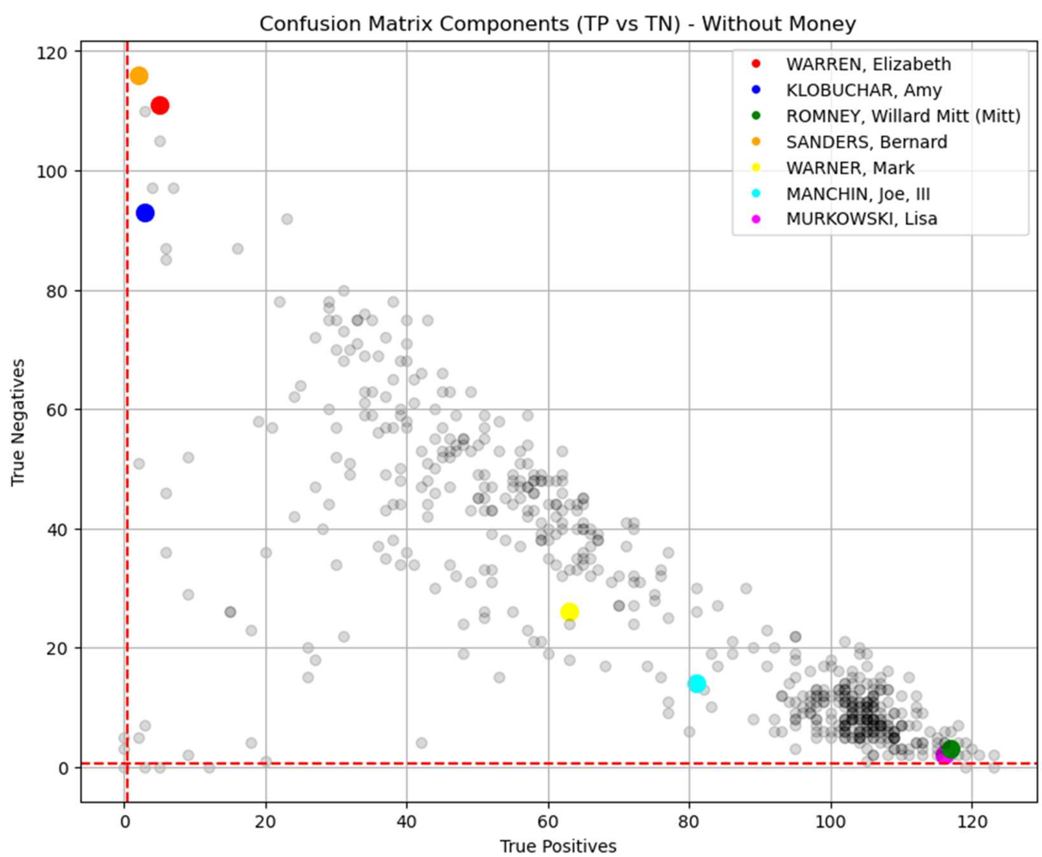
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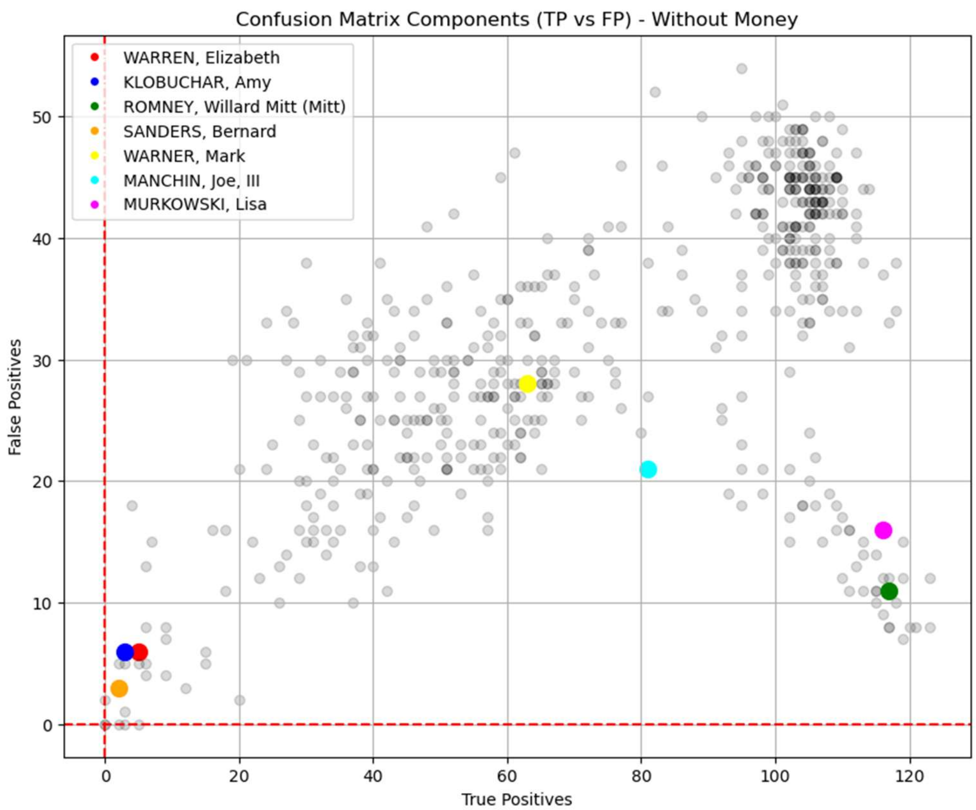
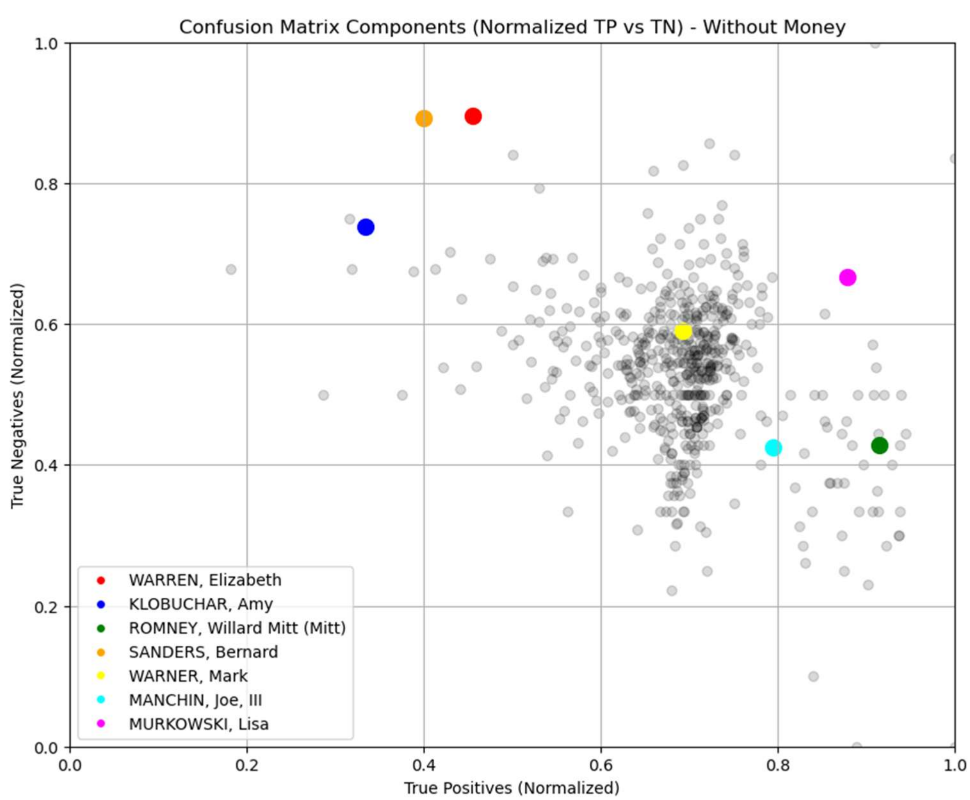
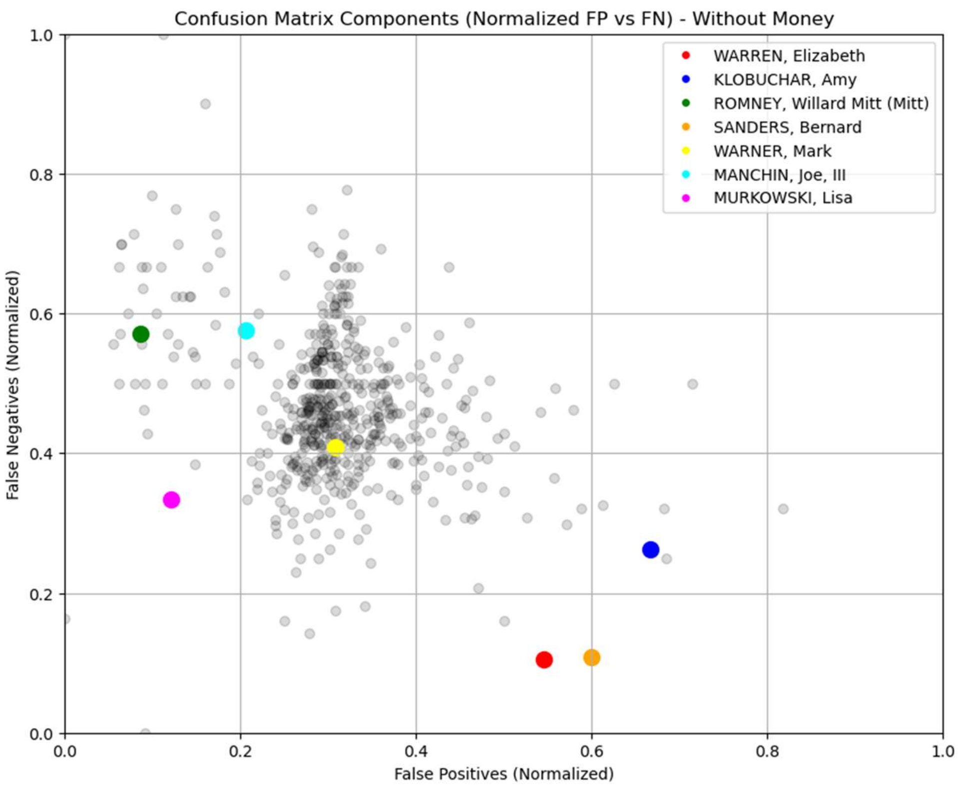


Normalized Graphs




Confusion Matrix Components (Normalized FP vs FN) - Without Money
WARREN, Elizabeth
KLOBUCHAR, Amy
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Conclusion